Carry-Sudoku & Slot-Sudoku in Reduced SHA-256: A Self-Contained ELI5 and ELI-Undergraduate Companion Guide

Author: Sebastian Wain (`srw@egont.xyz`) **Companion Pedagogical & Reference Document to:** *Practical Preimages for 19- and 20-Step Reduced SHA-256: Exact Algebraic Padding Inversion, Context Shaping, and Causal Bit-Slice Pruning* (`PAPER_19_20_PADDED_SHA256.pdf`)

Executive Overview: Why a "Sudoku" Guide to SHA-256?

Cryptographic hash functions like SHA-256 are designed to work like one-way blenders: you drop a message in at the top, turn the crank for 64 rounds of bitwise mixing and 32-bit addition, and get a 256-bit fingerprint at the bottom. Trying to run the blender backward—starting from a target fingerprint and finding a validly padded input message that produces it (*a preimage attack*)—looks impossible by brute force, because there are 2^{256} possible target states.

However, when SHA-256 is reduced to **19 steps (padded)** or **20 steps (padded)**, you don't have to guess the whole message at once, and you don't have to fight the blender head-on. Instead, you can treat the internal step-by-register grid of SHA-256 like a **giant interconnected Sudoku puzzle**:

1. **You pre-fill strategic "givens" (fixed cells) in the middle of the board** (`e9 = e10 = e15 = 0xffffffff` , `a5 = a6` , `a10 = a11`) so that nonlinear logic gates (Ch and Maj) turn into simple pass-through wires or constants (`[E8]` – `[E13]` , `[E22]`).
2. **You solve the padding byte (`0x80`) in $O(1)$ time and solve the arithmetic carry chains bit-slice by bit-slice from right to left (`Carry-Sudoku`)** (`[E18]` – `[E25]`), where knowing the rightmost bits of e_{13} immediately forces the carries and eliminates 99.74% of invalid branches before the heavy rotations are even computed.
3. **You invert the coupled schedule operator $Q_0(u) = \sigma_0(u) - u$ across 3 chained steps using a zero-table closed-form $\mathbb{F}_2$ inverse and a packed hash table (`Slot-Sudoku` / Cuckoo-Sudoku)** (`[E27]` – `[E37]`), eliminating memory bottlenecks on both Apple M3 CPUs and NVIDIA L4 GPUs.

This companion guide is organized so that **anyone**—from a curious reader to a cryptography undergraduate or systems engineer—can understand every single piece of the breakthrough:

- **Section 0 (Master Glossary & Complete `[E1]` – `[E45]` Index):** A 100% self-contained reference explaining every symbol, operator, C/CUDA variable, and numbered equation (`[E1]` through `[E45]`) in the main paper.
- **Section 1 — The Scalar a-Sequence Grid (`[E1]` – `[E7]`):** How the 8-register SHA-256 state collapses into a single 1D Sudoku strip $a_{-7}, \ldots, a_R$, and why FIPS 180-4 padding (`0x80` , zero fill, length `440` or `952`) locks the board (*ELI5* and *ELI-Undergraduate*).
- **Section 2 — Selector Saturation & Exact Context Shaping $D = 0$ (`[E8]` – `[E14]`):** How locking $e_9 = e_{10} =$ `0xffffffff` and forcing twin cells $a_5 = a_6$ ($D = a_5 \oplus a_6 = 0$) decouples the Left and Right halves of the cipher (*ELI5* and *ELI-Undergraduate*).
- **Section 3 — Exact $O(1)$ Algebraic Padding Inversion (`[E15]` – `[E22]`):** How to absorb $W_{15} = 440$ and $W_{14} = 0$ in closed form and solve $(W_{13} \bmod 256) =$ `0x80` in $O(1)$ time with zero trial-and-error (*ELI5* and *ELI-Undergraduate*).
- **Section 4 — `Carry-Sudoku` : 7-Stage Causal Bit-Slice Right-Context Solver (`[E23]` – `[E25]`):** How Theorem 1 splits $a_5 = a_6$ into a lower-triangular carry check independent of a_6, reducing 2^{24} candidates per a_{12} block to an av-

erage of $43{,}580$ node evaluations ($385\times$ reduction) with 100% completeness and reaching **`266.8 ctx/s`** on CPU and **`114,507.0 exact $D=0$ contexts/s`** on 1× NVIDIA L4 GPU (*ELI5* and *ELI-Undergraduate*).

- **Section 5 — `Slot-Sudoku` & Zero-Table Closed-Form $\mathbb{F}_2$ Inverse (`[E27]`–`[E37]`):** How Theorem 2 reconstructs the upper 16 bits in closed form (`reconstruct_u`), how 6 GiB / 16 GiB / 32 GiB `Slot-Sudoku` tables work, and how every 19-step bridge candidate (`sol19`) is filtered in $O(1)$ via σ_0^{-1} to find the first 20-step padded preimages (*ELI5* and *ELI-Undergraduate*).
- **Section 6 — GPU Warp-Sudoku, Exact-T_{16} Twin Coalescing & The Zero-a_3 Unpadded Bridge (`[E38]`–`[E45]`):** How our 16-thread CPU solver reaches **`6,242.8 sol19/s`** (**`191.1 node-hours`** per 20-step padded preimage on a 10-core Apple M4, or $2^{46.56}$ **scalar CPU reference compressions**), how supplementary GPU acceleration in a_1-slice sweeps on 1× NVIDIA L4 reaches **`53,847.5–73,560.5 sol19/s`** (`16.22–22.16 GPU-h` projected) on the 6 GiB EDF table and **`132,888.7 sol19/s`** (**`8.98 GPU-hours` projected** on 1× L4, or **`1.12 hours` across 8× L4 GPUs**) on genuine birthday-collided T_{16} context pairs, and how setting $a_3 = 0$ solves 20-step unpadded compression preimages in 21.1 minutes observed on CPU (*ELI5* and *ELI-Undergraduate*).
- **Section 7 — Scope, "Restricted" Preimages, Two-Block Plain-ASCII Preimages, and 15 Explored Sub-2^{32} Directions:** Why locking Sudoku cells doesn't throw away the solution ($440 \to 184 \to 74.6$–78.8 free bits), how a **2-block 119-byte message** lets you choose **any 64-byte plain-ASCII first block** in $O(1)$ while **Direction 15 (Two-Block Feed-Forward Locking)** explains why 2 blocks cannot reduce Round 20 below 2^{32}, and the taxonomy of **15 explored sub-2^{32} directions** (*ELI5* and *ELI-Undergraduate*).

Section 0: Master Self-Contained Glossary & Complete `[E1]`–`[E45]` Index

0.1 Core Symbols, Operators, and Code Variables

Symbol / Code Name	Name	Self-Contained Meaning in Plain English & Mathematics
$+, -$ (modulo 2^{32})	32-bit Modular Arithmetic	Addition and subtraction on 32-bit unsigned integers (`uint32_t` in C/CUDA), wrapping around modulo 2^{32}.
$\oplus, \wedge, \vee, \neg$	Bitwise Logic Operators	Bit-by-bit XOR (`^`), AND (`&`), OR (`|`), and bitwise NOT (`~`) across 32 bits ($b \in \{0, \dots, 31\}$).
$\mathrm{ROR}(x, n), \mathrm{SHR}(x, n)$	Right Rotate & Right Shift	Circular right rotation by n bits (`(x >> n) | (x << (32-n))`) and logical right shift (`x >> n`).
$\mathrm{Ch}(e, f, g)$	Bitwise "Choose" Gate	$(e \wedge f) \oplus (\neg e \wedge g)$. At each bit b, if $e_b = 1$, output f_b; if $e_b = 0$, output g_b. Setting $e =$ `0xffffffff` forces $\mathrm{Ch}(e, f, g) = f$, erasing g!
$\mathrm{Maj}(a, b, c)$	Bitwise "Majority" Gate	$(a \wedge b) \oplus (a \wedge c) \oplus (b \wedge c)$. Outputs the bit (`0` or `1`) that appears at least twice. If two inputs are equal ($a = b$), $\mathrm{Maj}(a, a, c) = a$, erasing c!
$\Sigma_0(x), \Sigma_1(x)$	State Mixing Bijections	Invertible $\mathbb{F}_2$-linear state mixers: $\Sigma_0(x) = \mathrm{ROR}(x, 2) \oplus \mathrm{ROR}(x, 13) \oplus \mathrm{ROR}(x, 22)$ and $\Sigma_1(x) = \mathrm{ROR}(x, 6) \oplus \mathrm{ROR}(x, 11) \oplus \mathrm{ROR}(x, 25)$.
$\sigma_0(x), \sigma_1(x)$	Schedule Mixing Maps	$\mathbb{F}_2$-linear message mixers: $\sigma_0(x) = \mathrm{ROR}(x, 7) \oplus \mathrm{ROR}(x, 18) \oplus \mathrm{SHR}(x, 3)$ (invertible over $\mathbb{F}_2^{32}$!) and $\sigma_1(x) = \mathrm{ROR}(x, 17) \oplus \mathrm{ROR}(x, 19) \oplus \mathrm{SHR}(x, 10)$.
$Q_0(u), Q_1(u)$	Coupled Schedule Operators	$Q_0(u) := \sigma_0(u) - u \pmod{2^{32}}$ and $Q_1(u) := \sigma_1(u) + u \pmod{2^{32}}$. Used to invert the message schedule across steps 16, 17, 18.

Symbol / Code Name	Name	Self-Contained Meaning in Plain English & Mathematics
$a_{-7}, \ldots, a_R$	Scalar a-Sequence	Because 6 of the 8 SHA-256 registers simply shift each round, the entire 8-register trajectory is uniquely determined by the single sequence of a_s words (`[E5]` – `[E7]`).
$D_{\text{shape}} = a_5 \oplus a_6$	Context Shaping XOR	Measures how many bits differ between a_5 and a_6. Exact shaping sets $D_{\text{shape}} = 0 \iff a_5 = a_6$ (`[E11]`).
$M_3 = \neg(a_3 \oplus a_5)$	Level-3 Majority Mask	The bitmask of positions where a_3 and a_5 agree. On these bits, a_4 must equal a_5 so that $\text{Maj}(a_5, a_4, a_3) = a_4$ (`[E13]`).
$D = a_8 \oplus a_7$	Carry-Sudoku Difference Mask	Inside Theorem 1 (`[E23]`), $D = a_8 \oplus a_7$ separates bits where $\text{Maj}(a_8, a_7, a_6)$ depends on a_6 ($D_b = 1$) from bits where it equals $a_{8,b}$ ($D_b = 0$).

0.2 Complete Cross-Reference of Equations `[E1]` – `[E45]` in `PAPER_19_20_PADDED_SHA256.pdf`

- `[E1]` **(FIPS 180-4 Single-Block 55-Byte Padding):** Fixes $W_{15} = 440$ (`0x000001b8`), $W_{14} = 0$ (`0x00000000`), and $W_{13} \bmod 256 = \texttt{0x80}$ (72 locked padding bits; or $W_{15} = 952 = \texttt{0x000003b8}$ for a 2-block 119-byte padded message).
- `[E2]` **(SHA-256 Boolean and Linear Functions):** Defines Ch, Maj, Σ_0, Σ_1, and σ_0, σ_1.
- `[E3]` **(Message Schedule Expansion):** $W_t = \sigma_1(W_{t-2}) + W_{t-7} + \sigma_0(W_{t-15}) + W_{t-16}$ for $t \geq 16$.
- `[E4]` **(Coupled Schedule Operator Q_0):** Defines $Q_0(u) = \sigma_0(u) - u \pmod{2^{32}}$.
- `[E5]` **(Initial Boundary Conditions):** Fixes $(a_0, a_{-1}, a_{-2}, a_{-3})$ and $(e_0, e_{-1}, e_{-2}, e_{-3})$ from IV.
- `[E6]` **(Terminal Boundary Conditions):** Fixes $a_{12} \ldots a_{19}$ (for $R = 19$) or $a_{13} \ldots a_{20}$ (for $R = 20$) from $H^* - \text{IV}$.
- `[E7]` **(Scalar a-Sequence Recovery):** Expresses every e_s as a deterministic function of 5 consecutive a-words ($a_{s-4} \ldots a_s$) and every W_t via $e_{t+1}, e_t, e_{t-1}, e_{t-2}, e_{t-3}, a_{t-3}$ (spanning 9 consecutive a-words $a_{t-7} \ldots a_{t+1}$ when fully expanded in a), leaving only $a_1 \ldots a_{12}$ free for $R = 20$ ($a_1 \ldots a_{11}$ for $R = 19$).
- `[E8]` **(Selector Saturation):** Pins $e_9 = \texttt{0xffffffff}$ and $e_{10} = \texttt{0xffffffff}$ to block e_7, e_8 from leaking rightward through Ch.
- `[E9]` **(Right-Only Expressions for a_5, a_6):** Expresses a_5 and a_6 purely in terms of the Right words (a_7, a_8, a_9, a_{10}).
- `[E10]` **(Collapsed Left Dependence of $W_8 \ldots W_{12}$):** Shows how $W_8, W_9, W_{10}, W_{11}, W_{12}$ depend on (a_1, a_2, a_3, a_4): $W_8 = c_8 - a_1 - e_5(a_1, \ldots, a_5)$, $W_9 = c_9 - a_2 - \big(a_4 - \text{Maj}(a_5, a_4, a_3)\big)$, $W_{10} = c_{10} - a_3 - \big(a_5 - \text{Maj}(a_6, a_5, a_4)\big)$, $W_{11} = c_{11} - a_4$, and $W_{12} = c_{12}$ (Right-only).
- `[E11]` – `[E12]` **(Exact Context Shaping $a_5 = a_6$):** Setting $D_{\text{shape}} = a_5 \oplus a_6 = 0$ (`[E11]`) forces $\text{Maj}(a_6, a_5, a_4) = a_5$ (`[E12]`), making $W_{10} = c_{10} - a_3$ 100% free of a_4 leakage.
- `[E13]` **(Majority Cancellation in W_9):** Enforces $\text{Maj}(a_5, a_4, a_3) = a_4 \iff ((a_4 \oplus a_5) \wedge M_3) = 0$ with $M_3 = \neg(a_3 \oplus a_5)$, making $W_9 = c_9 - a_2$ 100% free of a_3, a_4 leakage.
- `[E14]` **(Three Chained Q_0-Inversions):** Rewrites the schedule equations at $t = 16, 17, 18$ as Level 1 ($Q_0(W_1) = T_{16} + F_1(a_1)$), Level 2 ($Q_0(W_2) = T_{17} + G_2(a_1, a_2) - W_1$), and Level 3 ($Q_0(W_3) = T_{18} + G_3(a_1, a_2, a_3) - W_2$).
- `[E15]` – `[E17]` **(Closed-Form Absorption of $W_{15} = 440, W_{14} = 0$):** Computes e_{16}, e_{15}, e_{14} (`[E15]`), solves e_{12}, e_{11} in closed form from e_{13} (`[E16]`), and recovers a_8, a_7 (`[E17]`).

- **[E18] – [E21] (Exact $O(1)$ Pivot Inversion of $W_{13} \bmod 256 = \texttt{0x80}$):** Uses the bit-disjointness of $\Sigma_1(e_{13}) \bmod 256$ on bit 18 and bits 25..31 ([E18] – [E20]) to pin those 8 bits in closed form ([E21]) for any choice of bits 0..17 and 19..24.
- **[E22] (Ch-Annihilation & Twin Majority Simplification):** Pins $e_{15} = \texttt{0xffffffff}$ and $a_{10} = a_{11}$, making e_{12} and a_8 constant across all 2^{24} choices of e_{13}.
- **[E23] (Theorem 1 — Carry-Sudoku Exact Equivalence):** Proves $a_5 = a_6 \iff (((S \oplus X) \wedge D) = 0) \wedge (((a_6 \oplus S) \wedge \neg D) = 0)$, where $S = X + \mathrm{Ch}(D, X, a_8)$ is 100% independent of a_6 and $\Sigma_0(a_9)$.
- **[E24] – [E25] (7-Stage Causal Bit-Slice Pruning):** Checks $((S \oplus X) \wedge D) = 0$ across 7 LSB-to-MSB bit-slices ([E24] – [E25]), reducing 2^{24} candidates per a_{12} block to an average of $43{,}580$ node evaluations ($385\times$ reduction) with 100% completeness. *(Note: equation index [E26] is retired so external references to [E27] – [E45] remain invariant.)*
- **[E27] – [E29] (Theorem 2 — Closed-Form $\mathbb{F}_2$ Inverse `reconstruct_u`):** Solves the upper 16 bits u_{hi} from (y, u_{lo}) in closed form with zero lookup tables.
- **[E30] – [E33] (`Slot-Sudoku` Table Designs):** Defines the 6.00 GiB 2-Tier Nested 4-Slot EDF + Cuckoo-Sudoku table ([E30] – [E31] , 82.98% coverage), the 16.00 GiB Direct-Indexed 2-Way Exact GPU table ([E32] , storing up to 2 valid roots per y, 89.79% unconstrained coverage, zero false checks), and the 32.00 GiB 4-Way Exact CPU table ([E33] , 99.59% unconstrained coverage; 99.57% when reserving `u_lo = 0` as an empty-slot sentinel).
- **[E34] – [E35] (Early Hamming Pruning & 64-Bit SWAR Level-3 Filter):** Prunes $\mathrm{popcount}(M_3) \le w_{\mathrm{thresh}}$ ([E34]) and tests 4 Level-3 slots simultaneously in one 64-bit register word ([E35]).
- **[E36] – [E37] (Terminal 20-Step Filter & σ_0^{-1} Bijection):** Filters the 19-step padded bridge candidate stream (`sol 19`) via $\sigma_0(W_4) + W_3 \equiv T_{19} \pmod{2^{32}}$ ([E36]) and proves $\det_{\mathbb{F}_2}(\sigma_0) = 1 \implies W_4 = \sigma_0^{-1}(T_{19} - W_3)$ ([E37]).
- **[E38] – [E41] (Zero-a_3 Bi-Directional Q_0/Q_1 Bridge for Unpadded 20-Step Compression):** Pins $a_3 = a_5 = a_6 = a_8 = a_9 = 0, a_{10} = \texttt{0xffffffff}$ ([E38]) to annihilate both majority gates ([E39]) and solves the 20-step compression function ([E40] – [E41]) in 21.1 minutes observed on 16 CPU cores (33.1 min expected).
- **[E42] (L2-Tile Range Pre-Filter):** For an L2-resident slice of `Q0_TABLE` covering $y_1 \in [\mathrm{Tile}_{\mathrm{lo}}, \mathrm{Tile}_{\mathrm{lo}} + \Delta_{\mathrm{tile}})$, tests $(T_{16} + F_1(a_1) - \mathrm{Tile}_{\mathrm{lo}}) \bmod 2^{32} < \Delta_{\mathrm{tile}}$ in registers before issuing any global memory read.
- **[E43] (Exact-T_{16} Twin Coalescing `twin2` & 4-Way Register-Prefetched Reservoir Sweep `single4`):** Pairs genuine contexts with birthday collisions $T_{16}^{(i)} = T_{16}^{(j)}$ (`twin2`) to share Level-1 lookups, or sweeps 4 distinct contexts simultaneously (`single4`) across the estimated universe of $U \approx 1.68 \times 10^7$ contexts.
- **[E44] (Low-16-Bit Hamming & Majority Pre-Filter Before `reconstruct_u`):** Rejects 59.82% (at $w_{\mathrm{lo,thresh}} = 7$) or 3.84% (at $w_{\mathrm{lo,thresh}} = 11$) of Level-2 candidates via $\mathrm{popcount}(M_3 \wedge \texttt{0xffff}) \le w_{\mathrm{lo,thresh}}$, and rejects $\approx 93\%$ ($\approx 15\times$ reduction) of Level-3 Slot-1/Slot-2 candidates via $((a_4 \oplus a_5) \wedge M_3 \wedge \texttt{0xffff}) = 0$ using only the low 16 bits u_{lo} before running `reconstruct_u` .
- **[E45] (4-Round Crossed Carry-Sudoku Spine $a_2 = a_3 = a_4 = a_5$):** Evaluates 4 consecutive majority-collapsed rounds (`a2 = a3 = a4 = a5`) directly in GPU registers with zero DRAM tables (`9,617.2M` specialized crossed-spine evaluations/s on 1× L4 GPU).

Section 1: The Scalar a-Sequence Grid (`[E1]` – `[E7]`)

1.1 Explain Like I'm 5 (ELI5): Turning an 8-Lane Highway into a Single Strip of Paper

If you look at a textbook diagram of SHA-256, it looks terrifying: there are **8 parallel lanes of numbers** (a, b, c, d, e, f, g, h) scrambling into each other for 20 steps, plus 16 message slots ($W_0 \dots W_{15}$) that expand into 4 recycled slots ($W_{16} \dots W_{19}$).

How do we turn that 8-lane mess into a clean Sudoku strip? Look closely at what happens to the 8 lanes in each step:

- Lane b is just a copy of lane a from 1 step ago! Lane c is lane a from 2 steps ago! Lane d is lane a from 3 steps ago!
- Similarly, lanes f, g, h are just copies of lane e from 1, 2, and 3 steps ago!
- Even better: lane e itself is completely determined by lane a (`[E7]`)!

That means **there aren't really 8 independent lanes at all**—there is only **one single strip of numbers**: $a_{-7}, a_{-6}, \dots, a_{20}$!

- The top 8 cells of the strip ($a_{-7} \dots a_0$) are pre-filled "givens" printed on the board by the SHA-256 starting value IV (`[E5]`).
- The bottom 8 cells of the strip ($a_{13} \dots a_{20}$ for 20 rounds, or $a_{12} \dots a_{19}$ for 19 rounds) are pre-filled "givens" locked by the target hash H^* (`[E6]`).
- **The entire 20-round SHA-256 puzzle comes down to filling in just 12 empty cells in the middle of the strip:** $a_1, a_2, \dots, a_{12}$ (or 11 cells $a_1 \dots a_{11}$ for 19 rounds) so that the padding rules (`[E1]`) and the 4 recycled message rules (`[E3]`) hold!

1.2 Explain Like I'm an Undergraduate (ELIU): Derivation of the Scalar a-Sequence (`[E5]` – `[E7]`)

In SHA-256, the step update at round $t \in \{0, \dots, R-1\}$ computes two new words:

$$T_{1,t} = h_t + \Sigma_1(e_t) + \mathrm{Ch}(e_t, f_t, g_t) + K_t + W_t, \qquad T_{2,t} = \Sigma_0(a_t) + \mathrm{Maj}(a_t, b_t, c_t),$$

and sets $a_{t+1} = T_{1,t} + T_{2,t}$ and $e_{t+1} = d_t + T_{1,t}$, while shifting $b_{t+1} = a_t, c_{t+1} = b_t, d_{t+1} = c_t$ and $f_{t+1} = e_t, g_{t+1} = f_t, h_{t+1} = g_t$. Subtracting $T_{1,t} = a_{t+1} - T_{2,t}$ from $e_{t+1} = d_t + T_{1,t}$ eliminates $T_{1,t}$ and W_t completely, giving the first equation of `[E7]` (with index $s = t+1$ and $d_t = a_{s-4}$):

$$e_s = a_s - \Sigma_0(a_{s-1}) - \mathrm{Maj}(a_{s-1}, a_{s-2}, a_{s-3}) + a_{s-4}.$$

Once e_s is known for all s, solving $e_{t+1} = d_t + h_t + \Sigma_1(e_t) + \mathrm{Ch}(e_t, e_{t-1}, e_{t-2}) + K_t + W_t$ for W_t gives the second equation of `[E7]` (where $d_t = a_{t-3}$ and $h_t = e_{t-3}$):

$$W_t = e_{t+1} - a_{t-3} - e_{t-3} - \Sigma_1(e_t) - \mathrm{Ch}(e_t, e_{t-1}, e_{t-2}) - K_t.$$

Therefore, once the boundary cells $a_{-7} \dots a_0$ (`[E5]`) and $a_{R-7} \dots a_R$ (`[E6]`) are fixed, **any** choice of the 12 interior words $(a_1, \dots, a_{12})$ defines a valid 20-step SHA-256 computation hashing to H^*. We only need $(a_1, \dots, a_{12})$ to satisfy:

1. The **72-bit FIPS 180-4 padding constraint (`[E1]`)**: $W_{15} = 440$, $W_{14} = 0$, and $W_{13} \bmod 256 = \texttt{0x80}$ (72 bit constraints).
2. The **4 message schedule recurrence equations (`[E3]`)** for $t = 16, 17, 18, 19$ ($4 \times 32 = 128$ bit constraints).

Section 2: Selector Saturation & Exact Context Shaping $D = 0$ (`[E8]` – `[E14]`)

2.1 Explain Like I'm 5 (ELI5): Building a Soundproof Wall in the Middle of the Board

We have 12 empty cells ($a_1 \dots a_{12}$) to fill in:

- Think of the first 4 cells (a_1, a_2, a_3, a_4) as **Team Left**.
- Think of the last 8 cells $(a_5, a_6, a_7, a_8, a_9, a_{10}, a_{11}, a_{12})$ as **Team Right**.

Why can't Team Right just solve the padding slots (W_{13}, W_{14}, W_{15}) first, and then let Team Left solve the recycled slots ($W_{16}, W_{17}, W_{18}, W_{19}$) afterward? Because normally, whenever Team Left changes (a_1, a_2, a_3, a_4), that change ripples rightward through e_5, e_6, e_7, e_8 and scrambles the middle slots $W_9, W_{10}, W_{11}, W_{12}$ and the padding slots W_{13}, W_{14}, W_{15}!

How do we build a **soundproof wall** between Team Left and Team Right?

1. **Lock the Railroad Switches (`[E8]`):** We set $e_9 = \texttt{0xffffffff}$ and $e_{10} = \texttt{0xffffffff}$. Remember how the Choose gate $\mathrm{Ch}(e, f, g)$ works: when e is all `1`s, it picks f and **disconnects the track from** g! Locking e_9 and e_{10} to all `1`s disconnects e_7 and e_8, stopping Team Left's ripples dead in their tracks so $W_{12} \dots W_{15}$ depend only on Team Right and $W_{11} = c_{11} - a_4$!
2. **Make a_5 and a_6 Identical Twins (`[E11]` – `[E12]`):** Even with $e_9 = e_{10} = \texttt{0xffffffff}$, cell a_4 from Team Left tries to sneak into slot W_{10} through the Majority Vote gate $\mathrm{Maj}(a_6, a_5, a_4)$ (`[E10]`). How do we silence voter a_4? **We force voters a_5 and a_6 to cast the exact same ballot ($a_5 = a_6$, so $D_{\text{shape}} = a_5 \oplus a_6 = 0$)!** Since a_5 and a_6 agree on every bit, they win the 2-out-of-3 majority vote automatically ($\mathrm{Maj}(a_5, a_5, a_4) = a_5$), and a_4 is **100% locked out of $W_{10} = c_{10} - a_3$!**
3. **Cancel a_4 and a_3 Out of W_9 (`[E13]`):** In slot W_9 (`[E10]`), a_4 appears twice with opposite signs: $-a_4 + \mathrm{Maj}(a_5, a_4, a_3)$. Whenever a_4 wins the vote $\mathrm{Maj}(a_5, a_4, a_3) = a_4$ (`[E13]`), $-a_4 + a_4 = 0$ cancels out completely, leaving $W_9 = c_9 - a_2$!

2.2 Explain Like I'm an Undergraduate (ELIU): Formal Derivation of `[E8]` – `[E14]`

Let us trace equations `[E8]` through `[E14]` in detail:

1. **Selector Saturation (`[E8]` – `[E9]`):** Setting $e_9 = e_{10} = \texttt{0xffffffff}$ (`[E8]`) gives $\mathrm{Ch}(e_9, e_8, e_7) = e_8$, $\mathrm{Ch}(e_{10}, e_9, e_8) = \texttt{0xffffffff}$, and $\mathrm{Ch}(e_{11}, e_{10}, e_9) = \texttt{0xffffffff}$. Substituting $e_9 = e_{10} = \texttt{0xffffffff}$ into `[E7]` at $s = 9$ and $s = 10$ isolates a_5 and a_6 purely as functions of the Right variables (a_7, a_8, a_9, a_{10}) (`[E9]`):

$$
\begin{aligned}
a_5 &= \texttt{0xffffffff} - a_9 + \Sigma_0(a_8) + \mathrm{Maj}(a_8, a_7, a_6), \\
a_6 &= \texttt{0xffffffff} - a_{10} + \Sigma_0(a_9) + \mathrm{Maj}(a_9, a_8, a_7).
\end{aligned}
$$

2. **Exact Context Shaping (`[E10]` – `[E13]`):** Expanding $W_8, W_9, W_{10}, W_{11}, W_{12}$ via `[E7]` under `[E8]` yields `[E10]` :

$$
\begin{aligned}
W_8 &= c_8 - a_1 - e_5(a_1, \dots, a_5), & W_{11} &= c_{11} - a_4, \qquad W_{12} = c_{12}, \\
W_9 &= c_9 - a_2 - \big(a_4 - \mathrm{Maj}(a_5, a_4, a_3)\big), & W_{10} &= c_{10} - a_3 - \big(a_5 - \mathrm{Maj}(a_6, a_5, a_4)\big).
\end{aligned}
$$

- When we enforce **exact context shaping** $D_{\text{shape}} = a_5 \oplus a_6 = 0 \iff a_5 = a_6$ (`[E11]`), the identity $\text{Maj}(a_5, a_5, a_4) = a_5$ simplifies W_{10} to $W_{10} = c_{10} - a_3$ (`[E12]`) for **every** $a_4 \in \mathbb{Z}_{2^{32}}$.
- Simultaneously, in W_9, when does $\text{Maj}(a_5, a_4, a_3) = a_4$? On any bit position where a_3 and a_5 are *opposite* ($a_{3,b} \neq a_{5,b}$), the majority of $(a_{5,b}, a_{4,b}, a_{3,b})$ is automatically $a_{4,b}$! On bit positions where $a_{3,b} = a_{5,b}$ (marked by the mask $M_3 := \neg(a_3 \oplus a_5)$), we simply require $a_{4,b} = a_{5,b}$, i.e., $((a_4 \oplus a_5) \wedge M_3) = 0$ (`[E13]`). Then $W_9 = c_9 - a_2$ (`[E13]`), so W_9 depends only on a_2, W_{10} only on a_3, W_{11} only on a_4, and $W_{12}, \ldots, W_{19}$ are **100% independent of** (a_1, a_2, a_3, a_4)!

3. **The Three Chained Q_0-Inversions (`[E14]`):** Substituting $W_9 = c_9 - a_2$, $W_{10} = c_{10} - a_3$, $W_{11} = c_{11} - a_4$ and $W_t = a_{t+1} - G_t(a_1, \ldots, a_t)$ into `[E3]` for $t = 16, 17, 18$ produces the 3-level tree search (`[E14]`):

$$\begin{aligned}
\text{Level 1 } (t = 16): &\quad Q_0(W_1) = T_{16} + F_1(a_1), \\
\text{Level 2 } (t = 17): &\quad Q_0(W_2) = T_{17} + G_2(a_1, a_2) - W_1, \\
\text{Level 3 } (t = 18): &\quad Q_0(W_3) = T_{18} + G_3(a_1, a_2, a_3) - W_2.
\end{aligned}$$

Section 3: Exact $O(1)$ Algebraic Padding Inversion (`[E15]` – `[E22]`)

3.1 Explain Like I'm 5 (ELI5): Picking the 8 Secret Pins in the Padding Lock

Now Team Right $(a_5 \ldots a_{12})$ has to satisfy the postal inspector's three locked padding rules (`[E1]`):

1. $W_{15} = 440$ (message length),
2. $W_{14} = 0$ (zero block), and
3. The last byte of W_{13} must be the red STOP bead `0x80` ($W_{13} \bmod 256 = \texttt{0x80}$).

Here is how we unlock all three without guessing:

- First, once we pick a_{10}, a_{11}, a_{12} and a dial e_{13}, the equations for $W_{15} = 440$ and $W_{14} = 0$ (`[E16]`) immediately tell us the exact values of e_{12} and e_{11}!
- Second, what about the red STOP bead $W_{13} \bmod 256 = \texttt{0x80}$? Normally, random choices of e_{13} only hit `0x80` 1 time out of 256 ($1/256$).
- **The $O(1)$ Lockpick (`[E18]` – `[E21]`):** Look at how e_{13} is mixed inside W_{13}: it passes through $\Sigma_1(e_{13})$, which rotates e_{13} right by 6, 11, and 25 bits (`[E20]`). When we look at the lowest 8 bits of $\Sigma_1(e_{13})$, we discover a miracle: **bits 25..31 and bit 18 of e_{13} each touch exactly one bit of the low byte of W_{13}, and never touch the low byte of e_{13} itself!**
- So we can pick the other 24 bits of e_{13} however we want, and then **flip those 8 secret pins (bit 18 and bits 25..31) in a single step (`[E21]`)** so that $W_{13} \bmod 256 = \texttt{0x80}$ is guaranteed **100% of the time ($256\times$ speedup)**!

3.2 Explain Like I'm an Undergraduate (ELIU): Derivation of `[E15]` – `[E22]`

1. **Absorbing $W_{15} = 440$ and $W_{14} = 0$ (`[E15]` – `[E17]`):** Given (a_{10}, a_{11}, a_{12}) and the terminal boundary $a_{13} \ldots a_{20}$, `[E7]` immediately determines e_{16}, e_{15}, e_{14} (`[E15]`). Setting $W_{15} = 440$ and $W_{14} = 0$ in `[E7]` at $t = 15, 14$ yields `[E16]` :

$$e_{12} = (e_{16} - a_{12} - \Sigma_1(e_{15}) - K_{15} - 440) - \mathrm{Ch}(e_{15}, e_{14}, e_{13}),$$
$$e_{11} = (e_{15} - a_{11} - \Sigma_1(e_{14}) - K_{14}) - \mathrm{Ch}(e_{14}, e_{13}, e_{12}),$$

and `[E7]` at $s = 12, 11$ recovers a_8, a_7 (`[E17]`).

2. **Exact $O(1)$ Pivot Inversion of $W_{13} \bmod 256 =$ `0x80` (`[E18]` – `[E21]`):** Reducing W_{13} (`[E18]`) modulo 256 gives `[E19]` :

$$\Sigma_1(e_{13}) \equiv R_{\text{lo}} := \big(c_{W_{13}} - \mathrm{Ch}(e_{13,\text{lo}}, e_{12,\text{lo}}, e_{11,\text{lo}}) - \texttt{0x80}\big) \pmod{256}.$$

Notice that R_{lo} depends **only on the low 8 bits** $e_{13,\text{lo}} = e_{13} \bmod 256$! Meanwhile, for each bit $b \in \{0, \ldots, 7\}$, the b -th bit of $\Sigma_1(e_{13})$ is (`[E20]`):

$$\big(\Sigma_1(e_{13})\big)_b = (e_{13})_{b+6} \oplus (e_{13})_{b+11} \oplus (e_{13})_{(b+25) \bmod 32}.$$

- For $b \in \{0, \ldots, 6\}$, bit $(e_{13})_{b+25} \in \{25, \ldots, 31\}$ appears uniquely in row b and nowhere in bits $0..17$.
- For $b = 7$, bit $(e_{13})_{7+11} = (e_{13})_{18}$ appears uniquely in row 7 and nowhere in bits $0..17$. Hence, for **any** assignment to bits $0..17$ of e_{13} ($e_{13,\text{base}}$) and any assignment to bits $19..24$, setting bit 18 and bits $25..31$ via `[E21]` :

$$\Delta := \big(R_{\text{lo}} \oplus \Sigma_1(e_{13,\text{base}})\big) \bmod 256, \qquad e_{13,\text{fixed}} := e_{13,\text{base}} \mid \big((\Delta \wedge \texttt{0x7f}) \ll 25\big) \mid \big(((\Delta \gg 7) \wedge 1) \ll 18\big),$$

satisfies $(W_{13} \bmod 256) =$ `0x80` with **probability** 1!

3. **Ch-Annihilation & Twin Majority (`[E22]`):** By choosing a_{12} so that $e_{15} =$ `0xffffffff` and setting $a_{10} = a_{11}$ (`[E22]`), we get $\mathrm{Ch}(e_{15}, e_{14}, e_{13}) = e_{14}$ and $\mathrm{Maj}(a_{11}, a_{10}, a_9) = a_{11}$. Thus e_{12} **and** a_8 **become 100% constant across all** 2^{24} **choices of** e_{13}!

Section 4: `Carry-Sudoku` — 7-Stage Causal Bit-Slice Right-Context Solver (`[E23]` – `[E25]`)

4.1 Explain Like I'm 5 (ELI5): Solving Addition from Right to Left Without Knowing a_6

To build a valid Right Context, we need to find values of e_{13} where a_5 and a_6 are identical twins ($a_5 = a_6$, `[E11]`). Look at the formula for a_5 (`[E9]`):

$$a_5 = X + \mathrm{Maj}(a_8, a_7, a_6), \qquad \text{where } X = \texttt{0xffffffff} - a_9 + \Sigma_0(a_8).$$

Normally, to check if $a_5 = a_6$, you would have to compute all 32 bits of a_9, rotate a_9 through $\Sigma_0(a_9)$ to get a_6, and then add $X + \mathrm{Maj}(a_8, a_7, a_6)$ to get a_5. That means you can't check *anything* until all 32 bits of e_{13} are chosen!

The `Carry-Sudoku` Breakthrough (Theorem 1, `[E23]`): What if we **already assume** a_6 is going to equal a_5, and plug X in place of a_6 inside the Majority gate? Call that test sum $S = X + \mathrm{Ch}(D, X, a_8)$, where $D = a_8 \oplus a_7$ (`[E23]`). Notice two mind-blowing facts about S:

1. S **does NOT use** a_6 **or** $\Sigma_0(a_9)$ **at all!**
2. **Every bit where** a_8 **and** a_7 **differ (**$D_b = 1$**),** S_b **MUST equal** X_b **(`((S ^ X) & D) == 0` , `[E24]`)!** Why? Because on any bit where $a_8 \neq a_7$, $\mathrm{Maj}(a_8, a_7, a_6)$ equals a_6, so $a_5 = X + a_6 + \text{carry}$ can only equal a_6 if $X + \text{carry} = 0 \pmod 2$, which means S and X must match on bit b!

3. Since addition carries only flow from **right to left (bit 0 $\rightarrow$ bit 31)**, the lowest k bits of S, X, and D depend **only on the lowest k bits of e_{13}**!

So just like Sudoku, we fill in e_{13} **3 to 4 bits at a time from right to left across 7 causal stages (`[E25]`)**, throwing away any chunk where `((S ^ X) & D)` is non-zero! Across the 2^{24} candidate values of e_{13} per a_{12} block, the 7 stages evaluate an average of only $43{,}580$ **tree nodes ($385\times$ reduction, pruning 99.74% of the search space)** before ever computing $\Sigma_0(a_9)$—with **zero false negatives (100% completeness)**!

4.2 Explain Like I'm an Undergraduate (ELIU): Proof of Theorem 1 & The 7-Stage Pipeline (`[E23]` – `[E25]`)

Let us review **Theorem 1 (`[E23]`)**: Define $X := \texttt{0xffffffff} - a_9 + \Sigma_0(a_8)$, $D := a_8 \oplus a_7$, and $S := X + \mathrm{Ch}(D, X, a_8) \pmod{2^{32}}$. Then:

$$a_5 = a_6 \iff \Big(((S \oplus X) \wedge D) = 0\Big) \wedge \Big(((a_6 \oplus S) \wedge \neg D) = 0\Big).$$

- **Why does `((S ^ X) & D) == 0` prune causally (`[E24]` – `[E25]`)?** Because a_8 is constant (`[E22]`), $a_9 = e_{13} - c_{e_{13}}$, and $a_7 = c_{a_7,\text{full}} - (e_{14} \wedge e_{13}) + ((a_{10} \oplus a_8) \wedge a_9)$, every operation in X, a_7, D, and $S = X + \mathrm{Ch}(D, X, a_8)$ is **strictly causal (lower-triangular)**: bits $0..k$ of S, X, D depend exclusively on bits $0..k$ of e_{13}! Thus `solve20_padded_fast.c` and `solve20_padded_cuda.cu` assign the 24 free bits of e_{13} across 7 causal stages (`[E25]`):
- **Stage 1 (bits 0..3):** checks mask `0xf` (16.0 nodes).
- **Stage 2 (bits 4..7):** checks mask `0xff` (80.4 nodes), fixing R_{lo} and $(e_{13})_{18} \oplus (e_{13})_{13}$.
- **Stage 3 (bits 8..11):** checks mask `0xfff` (412.9 nodes).
- **Stage 4 (bits 12..14 + exact bit 18):** once bit 13 is known, bit 18 is known! Checks mask `0x7fff` ($1{,}043.8$ nodes).
- **Stage 5 (bits 15..17, checked through bit 18):** checks 19-bit mask `0x7ffff` ($3{,}507.8$ nodes) and pins bits $25..31$ via `[E21]`.
- **Stage 6 (bits 19..21):** checks 22-bit mask `0x3fffff` ($8{,}845.4$ nodes).
- **Stage 7 (bits 22..24 + full 32-bit verification):** checks full 32-bit `((S ^ X) & D) == 0` ($29{,}673.5$ leaves; only $\approx 2{,}149$ survivors per a_{12} block evaluate $\Sigma_0(a_9)$ and $a_6 == a_5$). Across $2^{24} = 16{,}777{,}216$ candidates per a_{12} block, the 7 stages visit an average of $43{,}579.9$ **nodes ($385\times$ reduction)** with 100% completeness, reaching **`266.8 ctx/s`** on 16 CPU threads and **`114,507.0 ctx/s`** on 1× NVIDIA L4 GPU!

Section 5: `Slot-Sudoku` & Zero-Table Closed-Form $\mathbb{F}_2$ Inverse (`[E27]` – `[E37]`)

5.1 Explain Like I'm 5 (ELI5): The Closed-Form Origami Unfolder and the 4-Slot Pigeonholes

Now that Team Right can generate thousands of exact $D = 0$ padded contexts per second, Team Left sweeps $a_1 \in [0, 2^{32})$ and solves the three chained equations (`[E14]`):

- **Level 1:** Find W_1 such that $Q_0(W_1) = T_{16} + F_1(a_1)$.
- **Level 2:** Find W_2 such that $Q_0(W_2) = T_{17} + G_2(a_1, a_2) - W_1$.
- **Level 3:** Find W_3 such that $Q_0(W_3) = T_{18} + G_3(a_1, a_2, a_3) - W_2$.

Here we hit two huge engineering challenges—and solve both with **origami math (`Theorem 2`)** and `Slot-Sudoku`:

1. **Cut Memory in Half with the Closed-Form Origami Unfolder (`[E27]` – `[E29]`):** Storing a 32-bit number u in every table slot takes 4 bytes per slot. What if we only store the **bottom 16 bits (u_{lo})** in each slot (cutting table size in half to 2 bytes per slot!)? Previously, recovering the top 16 bits (u_{hi}) required looking up a 128 KiB table in memory 200 billion times per sweep, clogging the GPU's cache. In **Theorem 2 (`[E28]` – `[E29]`)**, we prove that you don't need any lookup table at all: because $\sigma_0(u_{\text{hi}} \ll 16)$ shifts bits in a neat triangle, **simple bit-shift/XOR operations (`reconstruct_u`) unfold the exact top 16 bits directly inside CPU/GPU registers**!
2. **Pack the Pigeonholes with `Slot-Sudoku` (`[E30]` – `[E33]`):** Some target values y have 0 solutions to $Q_0(u) = y$, while others have 1, 2, 3, or 4 solutions.

- In our **6.00 GiB `Slot-Sudoku` table (`[E30]` – `[E31]`)**, every group of 16 adjacent y's shares 12 slots arranged by Earliest-Deadline-First (EDF) sliding windows plus a Cuckoo-Sudoku 4th overflow slot, capturing **82.98%** of all preimages in just 6 GiB!
- In our **16.00 GiB Direct-Indexed 2-Way Exact GPU table (`[E32]`)**, every 32-bit y directly indexes its own `uint32_t` holding up to two exact 16-bit roots $(u_{\text{lo},1}, u_{\text{lo},2})$—giving **zero false-positive checks** across all 3 levels!
- And every survivor of Level 3 is a **19-step padded bridge candidate (`sol19`)**, which we test in 1 step (`[E36]` – `[E37]`) to find our **20-step padded preimage**!

5.2 Explain Like I'm an Undergraduate (ELIU): Theorem 2 (`reconstruct_u`), SWAR Filtering, & σ_0^{-1} (`[E27]` – `[E37]`)

1. **Proof of Theorem 2 — Closed-Form Upper-Half Inverse (`[E27]` – `[E29]`):** Given $y = Q_0(u) = \sigma_0(u) - u \pmod{2^{32}}$ and the low 16 bits $u_{\text{lo}} = u \bmod 2^{16}$, let $U = u_{\text{hi}} \ll 16$. Since $(u + y) \bmod 2^{17} = \sigma_0(u) \bmod 2^{17}$, bit 16 of $u = u_{\text{lo}} + U$ equals U_{16} (with no carry from u_{lo} into bit 16), and $\sigma_0(u) = \sigma_0(U) \oplus \sigma_0(u_{\text{lo}})$, we know bits $0..16$ of $\sigma_0(U) \oplus U$ (`[E27]`):

$$\sigma_0(U) \oplus U \equiv v := (u_{\text{lo}} + y) \oplus \sigma_0(u_{\text{lo}}) \pmod{2^{17}}.$$

Now write $U = L(T) := T \oplus (T \ll 11) \oplus (T \ll 15) \pmod{2^{32}}$ for an upper-half word T. Why this specific operator $L(T)$? Because inside $\sigma_0(U) = \text{ROR}(U, 7) \oplus \text{ROR}(U, 18) \oplus (U \gg 3)$, the term $\text{ROR}(U, 18) = (U \gg 18) \oplus (U \ll 14)$ interacts with $(U \gg 7)$ and $(U \gg 3)$ at shifts $+11$ and $+15$! Substituting $U = T \oplus (T \ll 11) \oplus (T \ll 15)$ cancels the two upper bands across bits $0..15$ and yields $v_{16} = T_{23} \oplus T_{19} \oplus T_{16} = v_5 \oplus v_1 \oplus T_{16}$ at bit 16 (since $U_{16} = T_{16}, U_{23} = T_{23} = v_5$, and $U_{19} = T_{19} = v_1$), leaving a direct closed-form expression for T (`[E28]`):

$$\begin{aligned} b_0 &= \big((v \ll 15) \oplus (v \ll 11) \oplus v\big) \wedge \texttt{0x00010000}, \\ b_{12} &= \big((v \ll 14) \oplus (v \ll 3)\big) \wedge \texttt{0x00060000}, \\ T &= b_0 \mid b_{12} \mid \big((v \ll 18) \wedge \texttt{0xfff80000}\big), \end{aligned}$$

followed by $u = u_{\text{lo}} \mid \big(T \oplus (T \ll 11) \oplus (T \ll 15)\big)$ (`[E29]`).

2. **64-Bit SWAR Pre-Reconstruction Filter (`[E34]` – `[E35]`):** At Level 3, before reconstructing u_{hi}, notice that the majority condition $((a_4 \oplus a_5) \wedge M_3) = 0$ (`[E13]`) on bits $0..14$ depends only on u_{lo}! Using SIMD-Within-A-Register (SWAR, `[E35]`), we test 4 candidate 16-bit slots packed in a 64-bit word `w` in 3 instructions, rejecting $> 99\%$ of quads without unpacking a single slot!

3. **Terminal 20-Step Bijection (`[E36]` – `[E37]`):** Every candidate passing Level 3 satisfies steps $0..18$ and all 72 padding bits—so it is a **schedule-consistent 19-step padded bridge candidate (`sol19`)**! To extend to **20 steps**, we only need step 19's schedule equation (`[E36]`):

$$\sigma_0(W_4) + W_3 \equiv T_{19} := W_{19} - \sigma_1(W_{17}) - W_{12} \pmod{2^{32}}.$$

Since σ_0 is a bijection over $\mathbb{F}_2^{32}$ ($\det_{\mathbb{F}_2}(\sigma_0) = 1$, `[E37]`), $W_4 = \sigma_0^{-1}((T_{19} - W_3) \bmod 2^{32})$ is uniquely determined by W_3 and T_{19}, yielding a 2^{-32} filter per `sol19` bridge candidate!

Section 6: GPU Warp-Sudoku, Exact-T_{16} Twin Coalescing & The Zero-a_3 Unpadded Bridge (`[E38]` – `[E45]`)

6.1 Explain Like I'm 5 (ELI5): Keeping All 32 GPU Lanes Busy + The Unpadded Shortcut

1. **Why GPUs Need `Warp-Sudoku` & Exact-T_{16} Twin Coalescing (`[E42]` – `[E44]`):** On an NVIDIA L4 GPU, threads work in locked teams of 32 (**Warps**). At Level 2, only a few threads in a warp find a valid a_3 with small Hamming weight (`[E34]`). If those few threads immediately run Level 3 while the other threads wait, most of the GPU sits idle!

- **Direct 2-Way Table & Warp-Aggregated Level-3 Work Queue (`[E32]`):** Every thread uses `__ballot_sync` to pack surviving Level-2 candidates into a shared queue until at least 32 candidates are ready, so all 32 lanes execute Level 3 at 100% SIMD utilization!
- **1.0M-Context GPU Reservoir & Exact-T_{16} Twin Coalescing (`twin2`) / 4-Way Sweep (`single4`) (`[E43]`):** By generating a large reservoir of exact $D = 0$ contexts at `114,507.0 ctx/s` and sorting by T_{16}, birthday pairs with $T_{16}^{(2m)} = T_{16}^{(2m+1)}$ (`twin2`) share their Level-1 table lookup with zero synthetic modification, while `single4` sweeps 4 distinct contexts simultaneously from the full reservoir $U \approx 1.68 \times 10^7$.
- **Pre-`ulo2` 16-Bit Lower-Triangular Pruning (`[E44]`):** At Level 3, checking the lower 16 bits of the majority condition (`((a4 ^ a5) & M3 & 0xffff) == 0`) rejects $\approx 93\%$ **($\approx 15\times$ reduction)** of invalid Level-3 Slot-1 and Slot-2 candidates *before* `reconstruct_u` (`[E29]`), reaching **`53,847.5–73,560.5` genuine `sol19/s`** (**`16.22–22.16 GPU-h`** projected) on the 6 GiB EDF table and **`132,888.7` genuine `sol19/s`** (**`8.98 GPU-hours` projected** per 20-step padded preimage on 1× L4, or **`1.12 hours` across 8× L4 GPUs**) in a_1-slice sweeps on genuine birthday-collided T_{16} context pairs!

2. **The Unpadded 20-Round Shortcut (`Zero-a3 Bridge` , `[E38]` – `[E41]`):** What if the postal inspector doesn't require padding (the unpadded 20-round compression function)? Then we can set cell $a_3 = 0$ along with $a_5 = a_6 = a_8 = a_9 = 0$ (`[E38]`). Since two zeros always win a 2-out-of-3 vote (`Maj(0, a4, 0) = 0` , `[E39]`), voter a_4 is defeated **100% of the time without any Hamming weight filter loss**, solving 20-step unpadded SHA-256 in **21.1 minutes observed on 16 CPU threads** (33.1 min expected, `[E41]`)!

6.2 Explain Like I'm an Undergraduate (ELIU): Summary of Genuine Unmutated Benchmarks

Following standard reduced-round SHA-256 complexity accounting, we report complexity in **scalar CPU reference 20-step compression equivalents (`CPU-ref SHA20-eq`)** anchored to the 16-thread scalar CPU reference implementation on our 10-core Apple M4 (1.5125×10^8 SHA20-eq/s, where 1 node-h $= 2^{38.99}$ CPU-ref SHA20-eq), and report GPU acceleration in direct GPU wall-clock hours (`GPU-h`) projected from a_1-slice sweep telemetry alongside `CPU-ref SHA20-eq` equivalents:

Solver / Architecture	Table Structure	Right-Context Speed ($D = 0$)	19-Step Bridge Rate (`sol19/s`)	`CPU-ref SHA20-eq` per `sol19`	Expected 20-Step Padded Time & `CPU-ref` Complexity (1 Device)	Expected Time on 8× L4 GPUs
Primary CPU (`solve20_padded_fast.c`, 16 threads)	**6.0 GiB 2-Tier Nested 4-Slot Sudoku**	`266.8 ctx/s`	`6,242.8 sol19/s`	`24,228` ($2^{14.56}$)	`191.1 node-h` ($2^{46.56}$ CPU-ref)	—
Cycle-0 CUDA Baseline (1× L4 GPU, a_1-slice)	6.0 GiB 2-Tier Sudoku ($B = 2$)	`7,580 ctx/s`	`53,847.5 sol19/s`	`2,808.9` ($2^{11.46}$)	`22.16 GPU-h` ($2^{43.46}$ CPU-ref)	`2.77 hours`
Upgraded 6 GiB CUDA (`solve20_padded_cuda.cu`, 1× L4 GPU, a_1-slice)	6.0 GiB + Zero-Table `[E29]` + $B = 8$	`7,580 ctx/s`	`73,560.5 sol19/s`	`2,056.1` ($2^{11.01}$)	`16.22 GPU-h` ($2^{43.01}$ CPU-ref)	`2.03 hours`
Exact-T_{16} Twin CUDA (`16 GiB Direct 2-Way`, 1× L4 GPU, a_1-slice)	**16.0 GiB Direct 2-Way Exact (`[E32]`)**	**`29,310–114,507 ctx/s`**	**`132,888.7 sol19/s`**	**`1,138.2`** ($2^{10.15}$)	**`8.98 GPU-h`** ($2^{42.15}$ CPU-ref)	**`1.12 hours`**
Zero-a_3 Bi-Directional Bridge CPU (`solve20_zero_a3.c`, 16 threads)	48.0 GiB Unpadded Q_0^{-1}/Q_1^{-1} (`[E38]` – `[E41]`)	Precomputed	`2,160,000 sol19/s` (unpadded)	`70.0` ($2^{6.13}$)	`33.1 min` exp. ($2^{38.13}$ CPU-ref); `21.1 min` obs. ($2^{37.48}$)	—

Section 7: Scope, "Restricted" Preimages, Two-Block Plain-ASCII Preimages, and 15 Explored Sub-2^{32} Directions

7.1 Explain Like I'm 5 (ELI5): The 440-Coin Budget, Plain-ASCII First Blocks, and Why Round 20 Takes 2^{32} Bridge Candidates

1. **Wait—By Pre-Filling Sudoku Cells ($e_9, e_{10}, a_5 = a_6, e_{15}, a_{10} = a_{11}$), Aren't We Throwing Away the Winning Answers?** Imagine you are looking for a winning lottery ticket in a giant stadium, and **a target hash has an average of 2^{184} ($\approx 2.4 \times 10^{55}$) different winning 55-byte messages**! Why so many? Because a 55-byte message has **440 free bits ("coins")**, and matching a 256-bit hash only costs **256 coins**—leaving you with **184 extra coins** to spend however you like!

- Locking the soundproof wall ($e_9 = e_{10} =$ `0xffffffff`) costs **64 coins** (120 coins left).
- Making $a_5 = a_6$ identical twins costs **32 coins** (88 coins left).
- Canceling a_4 in the majority vote costs 13.40 **coins** (13.28 coins from $(3/4)^{32}$ plus 0.12 coins from the $w \leq 14$ cutoff, leaving 74.60 **coins** $\implies$ $2^{74.60}$ **winning 20-round messages still left!**).
- Finally, turning on the Right-context turbocharger ($e_{15} =$ `0xffffffff`, $a_{10} = a_{11}$) costs **64 coins**, leaving $2^{10.60} \approx 1{,}552$ **unfiltered 20-step padded preimages** across an estimated universe of $U \approx 1.68 \times 10^7 \approx 2^{24.00}$ **valid Right contexts per target** (of which $2^{8.69} \approx 413$ are reachable in our 2-way table configuration across U, or $250 = 2^{7.97}$ when restricted to $U_{\text{filt}} \approx 1.014 \times 10^7$ with optional heuristic filtering since $U_{\text{filt}}/U \approx 60.5\%$)!
- Pairing contexts by exact 32-bit birthday collisions $T_{16}^{(i)} = T_{16}^{(j)}$ (`twin2`) or sweeping uniformly across the full reservoir U (`single4`) solves **all 16 padded 20-step targets (`100%` solve rate)**!

2. **How a Two-Block (119-Byte) Message Gives You a 64-Byte Plain-English First Block for Free!** In a 1-block (55-byte) message, our solver picks all 55 bytes algebraically, so the message looks like random computer bytes (`ec5 31306...`). What if we use a **2-block (119-byte) message** $M = B_1 \parallel B_2$?

- **Block 1 (B_1, bytes `0..63`) has ZERO padding rules!** You can pick **any 64-byte plain-English sentence you want** (for example: `"This first block is plain text; the second one was computed....."`) and run 20-step SHA-256 forward in $O(1)$ time to get the middle chaining state $V = H^{(1)}$.
- **Block 2 (B_2, bytes `64..118` + `0x80` + zeros + length `952 bits = 0x3b8`)** is then solved by our exact same GPU solver starting from $\mathrm{IV}' = V$ instead of IV!
- Better yet: even if someone gives a trick target like $H^* = \mathrm{IV}$ (where $H^* - \mathrm{IV} = 0$ in a 1-block attack), choosing B_1 makes $H^* - V \neq 0$ completely non-degenerate!

3. **Can Two Blocks Reduce Round 20 Below 2^{32}? (Direction 15: The Two-Block Feed-Forward Trap!)** At first glance, you might think: *"Wait! In a 2-block message, Block 1 gives us 512 brand-new free coins! Can't we use Block 1 to pay for Round 20 in Block 2 so Round 20 becomes $O(1)$?"* When we trace how Block 1 connects to Block 2, we discover **The Two-Block Feed-Forward Trap (Direction 15)**:

- Because of SHA-256's **Davies–Meyer feed-forward addition** at the end of Block 2 ($H^*_{20} = V + S^{(2)}_{20} \pmod{2^{32}}$), Block 1's output V is wired to **BOTH ends of Block 2 at the same time**:
- The **top** of Block 2 starts at $\mathrm{IV}' = V$, AND
- The **bottom** of Block 2 must end at $S^{(2)}_{20} = H^*_{20} - V \pmod{2^{32}}$!
- So if you change Block 1 to tweak V, **both the top and bottom of Block 2 move simultaneously!** And if you solve Block 2's Right context first (fixing $S^{(2)}_{20}$), that locks **all 256 bits** of $V = H^*_{20} - S^{(2)}_{20}$, which would force you to invert a 256-bit 20-step hash on Block 1 ($\Theta(2^{32})$) anyway!

7.2 Explain Like I'm an Undergraduate (ELIU): Formal Degree-of-Freedom Accounting & Taxonomy of 15 Explored Sub-2^{32} Directions

1. **Formal Degree-of-Freedom Accounting ($440 \to 184 \to 74.60 \to 10.60$ Bits):**

Algebraic / Structural Constraint	Equations	Degrees of Freedom Spent	Expected Preimages per 256-Bit Target ($R = 19$ or $R = 20$)	19-Step Bridge Candidates (`sol19`) per 20-Step Target
55-byte FIPS 180-4 padded block ($L = 440$)	`[E1]`	72.0 bits (of 512)	$2^{184.00}$	$2^{216.00}$
Selector saturation ($e_9 = e_{10} =$ `0xffffffff`)	`[E8]`	64.0 bits	$2^{120.00}$	$2^{152.00}$
Exact context shaping ($a_5 = a_6 \iff D_{\text{shape}} = 0$)	`[E11]`	32.0 bits	$2^{88.00}$	$2^{120.00}$
Majority cancellation ($\mathrm{Maj}(a_5, a_4, a_3) = a_4, w \leq 14$)	`[E13]`	13.40 bits ($13.28 + 0.12$)	$2^{74.60}$	$2^{106.60}$
Right-context turbocharger ($e_{15} =$ `0xffffffff`, $a_{10} = a_{11}$)	`[E22]`	64.0 bits	$2^{10.60} \approx 1{,}552$ ($2^{8.69} \approx 413$ in 2-way table)	$2^{42.60}$ ($2^{40.69}$ in 2-way table)

2. **Construction-Specific Design Explorations of 15 Candidate Sub-2^{32} Directions (`test_all_12_theories.c`, Cycle 10, and Direction 15):** To document why we retained our 3-level Left-tree architecture under realistic memory (≤ 64 GiB), we evaluated 15 candidate modifications (verified via model checks and surrogate probes in `test_all_12_theories.c`, Cycle 10, and Section 9.5; these represent construction-specific design analyses rather than universal impossibility proofs):

#	Candidate Sub-2^{32} Mechanism	Mathematical / Empirical Outcome Within Our Left-Tree Architecture	Observed Bottleneck
1	Shift saturation wall right ($e_{10} = e_{11} =$ `0xffffffff`)	W_9 depends on a_5; Level 1 (W_{16}) becomes circular in a_5	Infeasible in our tree
2	Backward Q_0^{-1} tree from $W_{19} \to W_{16}$	$W_4(a_1..a_5)$ couples all 4 Left words simultaneously	$\Theta(2^{32})$
3	Cross-context MitM on $T_{19} = \sigma_0(W_4) + W_3$	Left tree path depends on 96-bit tuple (T_{16}, T_{17}, T_{18})	$\Theta(2^{48})$ birthday
4	Partial bit-slice / LSB-first causal solver for $(a_1..a_4)$	$\sigma_0(u)$ rotates by $7, 18$ bits, coupling LSBs to MSBs at every level	$\Theta(2^{32})$
5	Differential / rotational invariant trails across Q_0^{-1} (surrogate 3-level probe)	Addition carries in $Q_0(u) = \sigma_0(u) - u$ destroy linear invariants in 3-level surrogate map	No bias observed
6	Linearizing Q_0 over $\mathbb{F}_2^{32}$ or $\mathbb{Z}_{2^{32}}$ (surrogate 3-level probe)	Empirical Jacobian across 3-level surrogate map has full rank ($32/32$ bits)	Full rank ($32/32$)
7	Colliding T_{16} and T_{17} simultaneously across Right contexts	Colliding 64 bits of (T_{16}, T_{17}) requires $\geq 2^{32}$ Right contexts	$\Theta(2^{32})$
8	Relaxed context shaping ($D_{\text{shape}} = a_5 \oplus a_6 \neq 0$, low weight)	Restores a_4 leakage into W_{10}, coupling Level 2 to Level 3	Slower than $D = 0$
9	Higher-order multi-slot Q_0^{-1} precomputation (W_1, W_2 joint table)	Joint table over (a_1, T_{16}, T_{17}) requires 2^{96} entries	$\gg 64$ GiB
10	Symbolic Gröbner basis / ANF elimination on $(a_1..a_4)$	Degree-31 carry polynomials across 3 coupled 32-bit additions	$\geq 2^{32}$
11	SAT / SMT CDCL & CryptoMiniSat bit-blasting on $(a_1..a_4 \mid \text{Right})$	XOR-addition chains induce exponential unit-propagation bottlenecks	$\gg 10^4\times$ slower
12	Neural / RL / heuristic priority-queue guidance on a_1	$\sigma_0^{-1}(T_{19} - W_3)$ exhibits avalanche uniformity ($r^2 < 10^{-6}$)	$\Theta(2^{32})$
13	**4-Round Crossed Carry-Sudoku Spine ($a_2 = a_3 = a_4 = a_5$, `[E 45]`)**	Collapses 4 majority gates (`0 GiB DRAM`, `9.62B evals/s` on L4), but spends 96 Left bits turning T_{16}, T_{17}, T_{18} into a 96-bit Right filter	Overconstrained (2^{-64}/ctx)
14	**5-Step Crossed Carry-Sudoku Spine ($a_1 = a_2 = a_3 = a_4 = a_5$)**	Spends 128 Left bits (`1 trial/ctx`); under constraint-count model, 128-bit schedule match needs 2^{128} Right contexts	Overconstrained (2^{-128}/ctx)
15	**Two-Block (119-Byte) Feed-Forward Attack ($M = B_1 \parallel B_2$)**	Davies–Meyer feed-forward $S_{20}^{(2)} = H_{20}^{*} - V \pmod{2^{32}}$ couples $V = \text{Compress}_{20}(\text{IV}, B_1)$ to both ends of Block 2 simultaneously	$\Theta(2^{32})$ (`8.98–16.22 GPU–h` projected on 1× L4 + plain-ASCII B_1)

Standalone Verification (`verify_all_witnesses.py`)

Run the zero-dependency Python verifier to check **all 46 distinct 20-step and 19-step witnesses across 48 checks (including the `'abc'` demo+benchmark overlap)** (17 20-step padded checks across 16 distinct witnesses, 17 19-step padded checks across 16 distinct witnesses including the 0^{256} exact $a_5 = a_6$ witness and all 14 benchmark targets, 13 distinct 20-step unpadded witnesses, and 1 Stage-2 completeness regression witness) and all internal Carry-Sudoku invariants:

```
python3 verify_all_witnesses.py
```

Disclosure & Attribution

Building upon K. S. Basra's selector-saturation and backward padding-absorption framework (`https://github.com/kamb-code/sha256-r19-preimage`), the algebraic discoveries (closed-form $O(1)$ W_{13} padding pivot inversion, Theorem 1 / Carry-Sudoku, Theorem 2 / closed-form $\mathbb{F}_2$ upper-half reconstruction, Slot-Sudoku tables, and the Zero-a_3 bi-directional bridge), C/CUDA solvers, and preimage witnesses in this work were discovered and developed using **Google DeepMind Gemini 4 Argon**.